

ECON 101H, Introduction to Economics

Recitation 6: Taxation

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Spring, 2026

Problem 1: Graphing Calculators

In the market for graphing calculators at a college town, demand and supply are given by

$$Q_D = 120 - P \quad \text{and} \quad Q_S = -240 + 3P.$$

The calculator market is perfectly competitive, with many different producers and consumers.

(a) What is the equilibrium quantity of calculators sold, and what is the total surplus?

(b) Next semester, the university announces that graphing calculators will be allowed on all exams. Also, Pennsylvania Devices Inc., an electronics producer, starts producing graphing calculators in December. If both of these changes happen at the same time, how will the price and quantity of graphing calculators change in equilibrium, assuming all else equal?

(c) If the University imposes a \$20 tax on the buyers of graphing calculators, how many calculators will be purchased?